



GE Medical Systems

Technical Publications

Direction 2402690

Revision 4

Signa[®] OpenSpeed EXCITE

Field Replaceable Units (FRU)

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GE Medical Systems

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REVISION HISTORY

REV	DATE	REASON FOR CHANGE
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0	Jul 20, '04	Initial Production Release
1	Feb 25, '05	Added HFO5 Information
2	Jun 24, '05	Miscellaneous Correction
3	Mar 09, '06	Miscellaneous Correction
4	Jul 07, '06	Miscellaneous Correction

SEND COMMENTS ABOUT THIS MANUAL TO:

GE MEDICAL SYSTEMS
MANAGER, SERVICE ENGINEERING
W-808
3200 N. GRANDVIEW BLVD.
WAUKESHA, WISCONSIN 53188
USA

This manual belongs to:

INTRODUCTION TO FIELD REPLACEABLE UNIT (FRU)

This direction is a guide to the FRU's.

Parts designated as a FRU are critical to maintain the safety and/or functionality of the system. FRU's are available on a Priority 1 basis from Service Parts Operation.

As a reference aid, the FRU's are listed under an upper-level part number where they are used. In most cases the upper-level part number is not a FRU. In some cases the parts listed under an upper-level assembly may have been nested in intermediate sub-assemblies which are not shown.

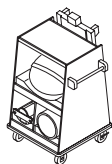
Where possible, FRU's which are part of Option Catalogs are shown under the area where they would be installed, rather than being listed in the "Coils & Options" section, Pages 31 & 32.

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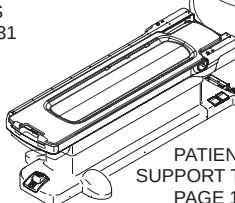
INTRODUCTION TO FIELD REPLACEABLE UNIT (FRU)PAGE 1COILS & OPTIONSPAGES 33 & 34

MAGNET
ENCLOSURE
ASSEMBLIES
PAGES 6 TO 9

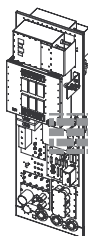
COVERS &
TOUCH-UP
PAINT
PAGE 5



SERVICE
TOOLS
PAGE 31



PATIENT
SUPPORT TABLE
PAGE 12



PENETRATION
PANEL
PAGE 18

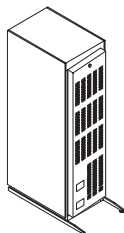
MAGNET ROOM**OPERATOR ROOM**

WORKSPACE SCIM
AND KEYBOARDS
PAGE 23

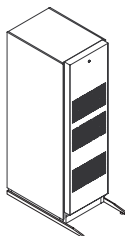
OPERATOR
WORKSPACE
PAGE 20



GOC
PAGE 21



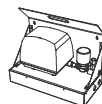
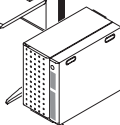
GRFD
CABINET
PAGE 25



SYSTEM
CONTROL
CABINET
PAGE 27



MAIN
DISCONNECT
PANEL
PAGE 25

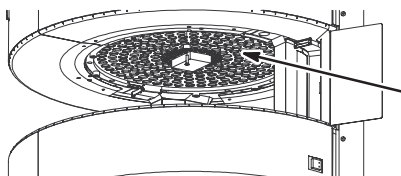


PATIENT
AIR SUPPLY
COMPRESSOR
PAGE 8

[illegible]

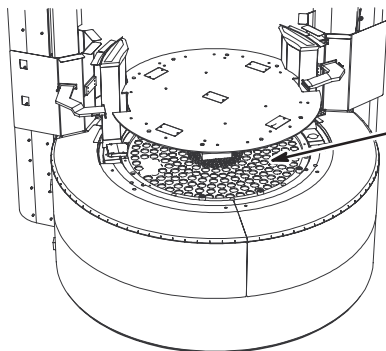
2183491-3	GEMS GRAY #1	0.6 OZ. Non-Aerosol
2183491	GEMS GRAY #2	0.6 OZ. Non-Aerosol
2183492	GEMS GRAY #2	4.5 OZ. Aerosol
2183491-2	GEMS GRAY #3	0.6 OZ. Non-Aerosol
2183492-2	GEMS GRAY #3	4.5 OZ. Aerosol
2180026	GEMS GRAY #4	0.6 OZ. Non-Aerosol



TOP GRADIENT COIL**5107866****TOP GRADIENT FRU**

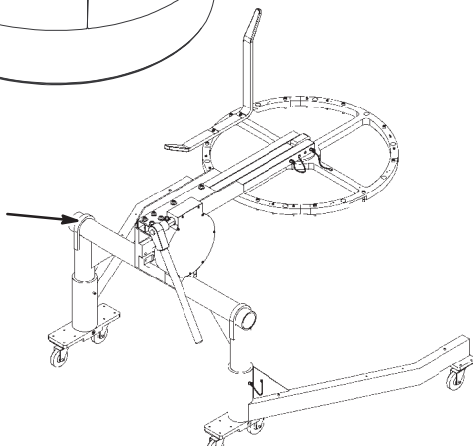
Includes:

- TOP GRADIENT COIL
- COIL CRATE (Blue)
- (10) 10mm BOLTS

BOTTOM GRADIENT COIL**5107866-2****BOTTOM GRADIENT FRU**

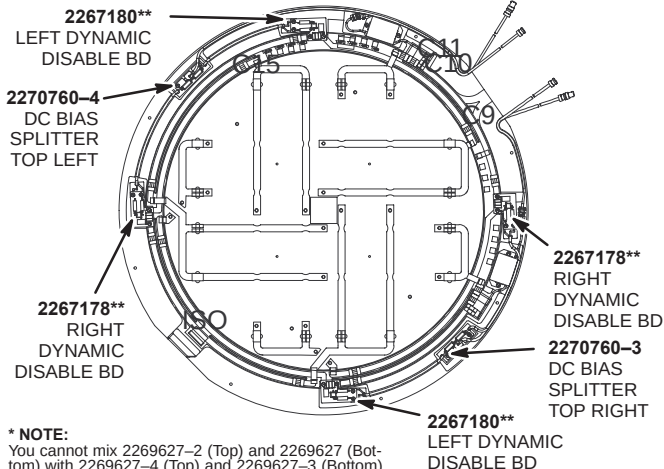
Includes:

- BOTTOM GRADIENT COIL
- COIL CRATE (Black)
- (10) 10mm BOLTS

2247491
GRADIENT
INSERTION
TOOL

REVISION 4

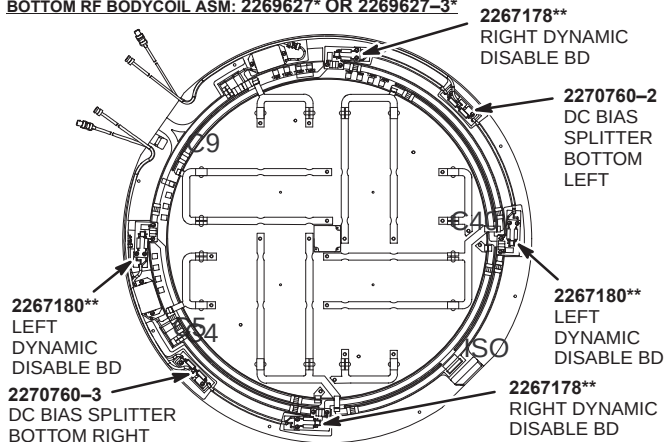
DIRECTION 2402690

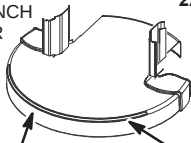
TOP RF BODYCOIL ASM: 2269627-2* OR 2269627-4**** NOTE:**

You cannot mix 2269627-2 (Top) and 2269627 (Bottom) with 2269627-4 (Top) and 2269627-3 (Bottom).

**** NOTE:**

2267178 and 2267180 may be used on 2269627-2 (Top), 2269627 (Bottom), 2269627-4 (Top), and 2269627-3 (Bottom).

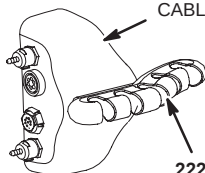
BOTTOM RF BODYCOIL ASM: 2269627* OR 2269627-3*

AIR DUCT COLLECTOR KIT: 2279600VIEWED FROM FRONT OF MAGNET**2276269-2**
EXHAUST RF TOPEXHAUST**2276269**
EXHAUST RF BOTTOM**2276268-2**
INPUT RF TOPINPUT**2276268**
INPUT RF BOTTOM**2276266**
RF COOLING
ELBOW
ADAPTER**PINCH SENSORS & TEMP SENSORS:****2270630**LEFT PINCH
SENSOR**2270630-2**RIGHT
PINCH
SENSOR

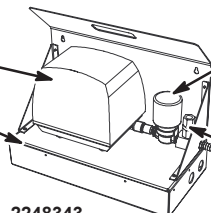
(UNDERNEATH)

2299536TEMP SENSOR
CABLE (RUN 749)**2299537**

TEMP SENSOR (TO RUN 749)

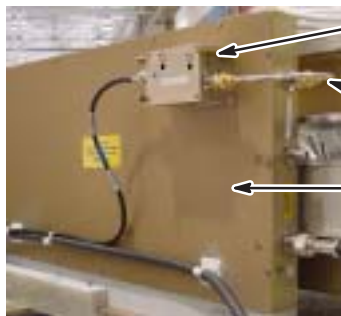
2278971
FRONT PINCH
SENSOR ASM**REMOTE PAC INTERFACE ASM:**
2227142 or 2227142-2**2221024**
PAC HANGER
CABLE I/F**2221023**
HANGER BAR**COMPRESSOR:****2248343-2**
COMPRESSOR**2248343-3**
SOLID STATE
RELAY (INSIDE)**2248343**

COMPRESSOR ASSEMBLY

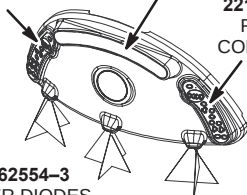
2248343-4
MUFFLER**2248343-5**
RELIEF
VALVE

REVISION 4

DIRECTION 2402690

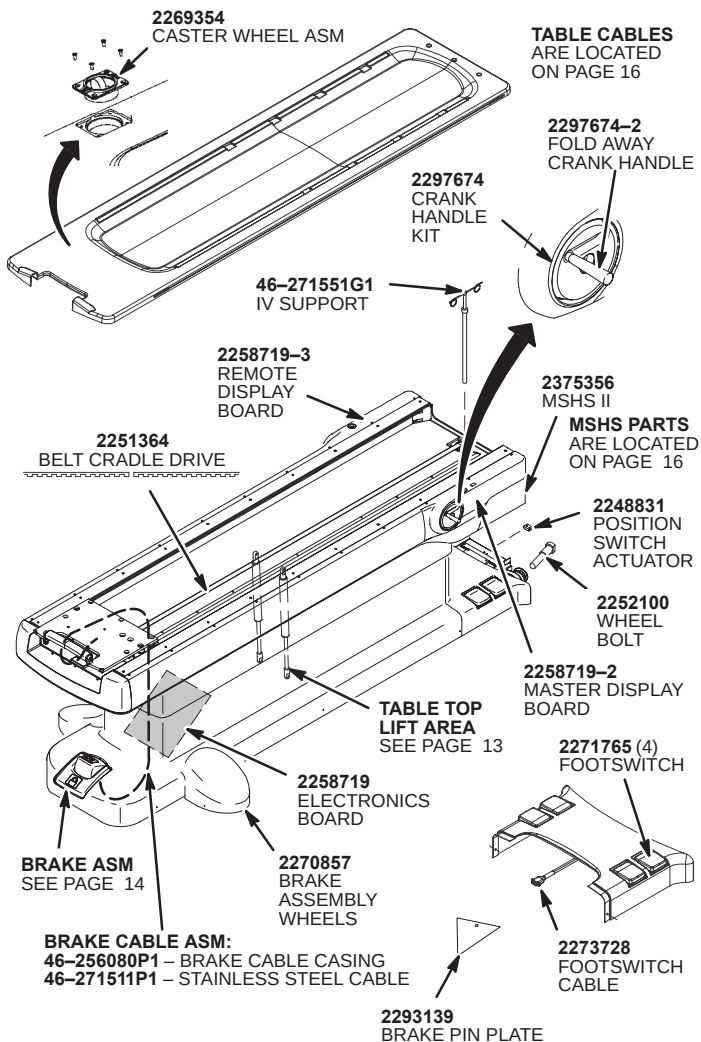
PAC/SRI:**46-321338G4**0.7T PREAMP W/MICROWAVE
ABSORBERS & DIODES**2364795-2**RUN 983 COAXIAL CABLE
MG3A3J2 TO MG3A3A5 IN**2244307**

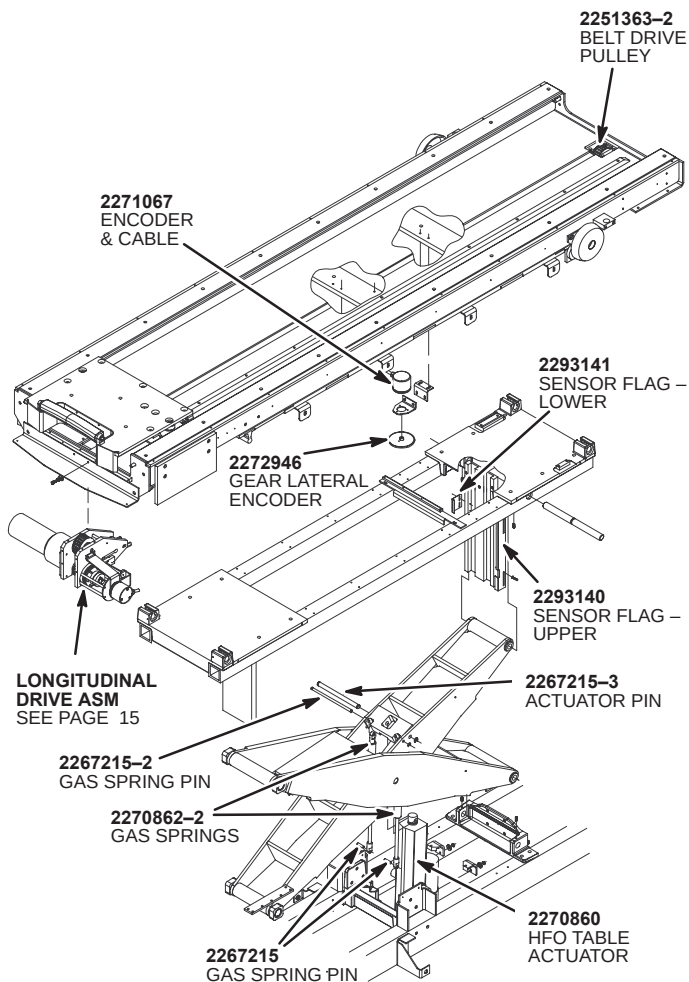
0.7T BODY HYBRID/TR ASM

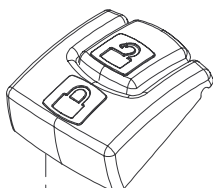
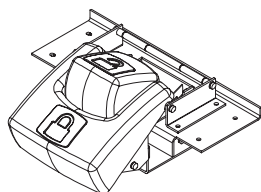
2305633-2PHYSIOLOGICAL ACQUISITION
CONTROLLER (PAC)**2255782**BODY HYBRID/TR
ASM & PREAMPFor HFO5~
5105270
SRI-3 ASMFor HFO4
2299497
SRI-2 ASM**2204384 OR**
46-264758G1
REMOTE
INTERCOM
BOARD**2284771**
DB15 FILTER**2284772**
DB25 FILTERDISPLAY/CONTROLS:**2215716-3**
LEFT
CONTROLS**2215716-4**
CENTER
DISPLAY**2215716-2**
RIGHT
CONTROLS**2162554-3**
LASER DIODESLAMPS:**2257458**
HALOGEN LAMP
12V20W
Q20MR16C/CG40DE
(Part of Comfort Modules
2216200 & 2216200-2)

HFO5 Related CABLES:**5123813** RUN 1261 Operator Display to SRI**5131886** RUN 749 Temp Sensor Cable**5131886** Upgrade Cable for Temp Sensor Cable

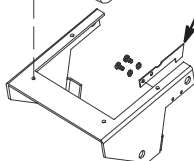
[illegible]







2252281
FLAG



2264412
ROLLER

2264413
ROLLER PIN

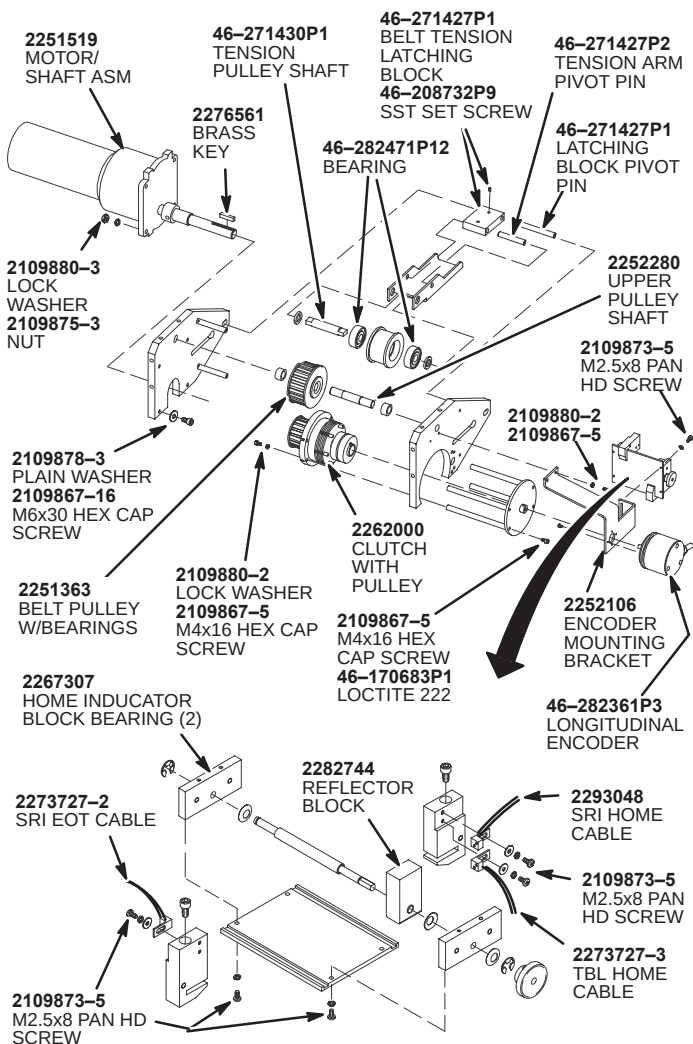
2264402
CHANNEL PIN

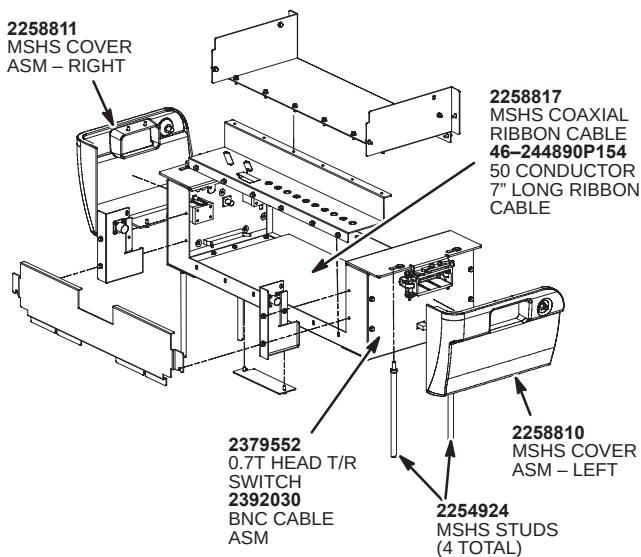
2269347
BELL CRANK PIN

2269349
BRAKE STOP
HOLDER

2269349
BRAKE STOP
HOLDER

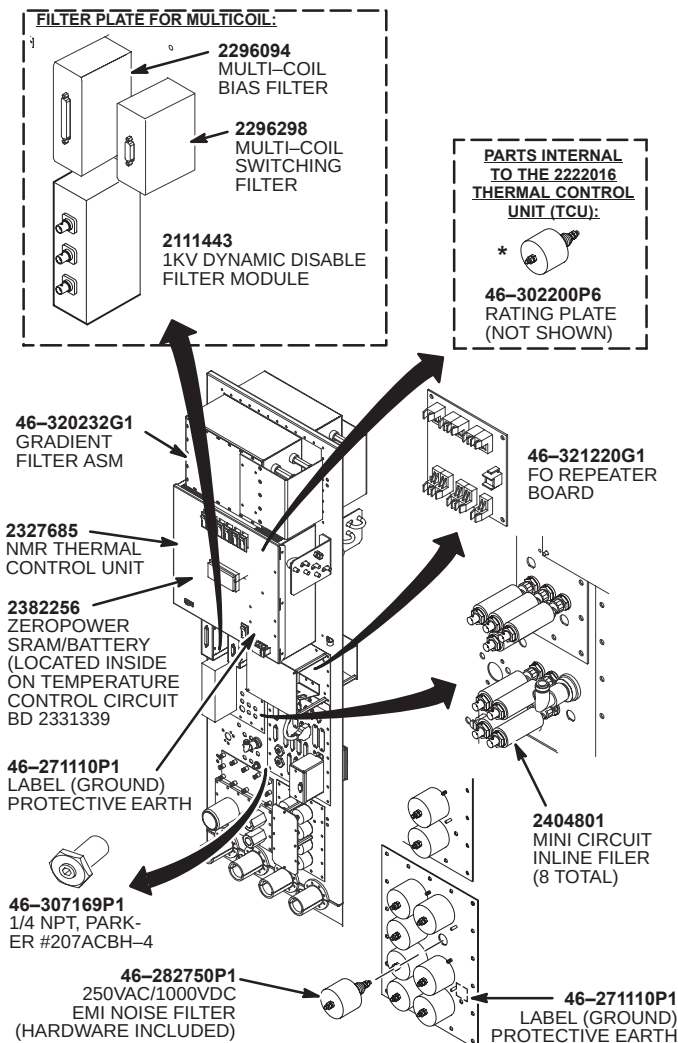
2269345
BRAKE STOP

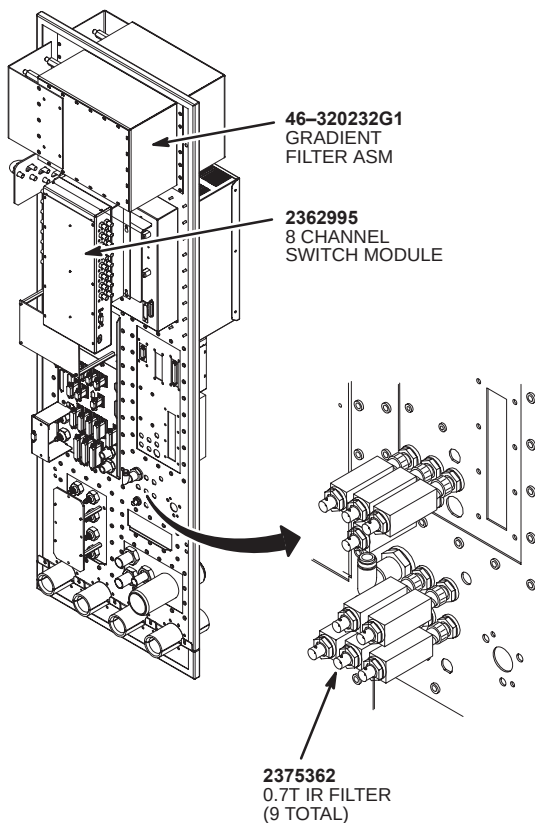


MSHS (Mousehouse):

CABLES:

46-317220P19	RUN 875
46-317220P20	RUN 876
46-317220P21	RUN 877
46-317220P22	RUN 878
2262696	RUN 881
2263200-30	RUN 882
2263200-32	RUN 883
2263200-33	RUN 884
2263200-31	RUN 885
2271496	RUN 886
46-320888G3	RUN 891
46-320888G4	RUN 892
2393987-3	RUN 1161
2393987-4	RUN 1162
2393987-5	RUN 1163
2393987	RUN 1164
2393987-2	RUN 1165
2384547	RUN 1167
2273042	FOOTSWITCH
2274468	DOWN LIMIT PHOTO SENSOR/CABLE ASM
2274468-2	UP LIMIT PHOTO SENSOR/CABLE ASM
2274468-3	SRI UP LIMIT PHOTO SENSOR/CABLE ASM
2274468-4	PIVOT LOCK PHOTO SENSOR/CABLE ASM
2263200-28	LAT DISPLAY MASTER - J2 TO MG3A5J3
2263200-29	LAT DISPLAY REMOTE - J1 TO LAT
2275282	TABLE VERTICAL MOTOR TO MG3A5J11





OPERATOR WORKSPACE TABLE:**For HFO5****5116657**SCSI TOWER with MOD and
DVD-RW**5116795**

Sony DVD-RW Internal Drive

2368521

Adapter: SCSI to IDE Bridge

2350404

Terminator, CENT50, ACTIVE

5127237

15 Inch LCD Desktop Monitor

or

2138599-2LCD DISPLAY (800 X 600)
W/POWER CORD,
VGA VIDEO CABLE,
& MOUNTING HARDWARE**For HFO4****2381744**

SCSI TOWER

2264509 SERIAL
CONVERTER BOX
W/ELECTRONICS**2138599-3**VGA VIDEO
CABLE**2349792***

or

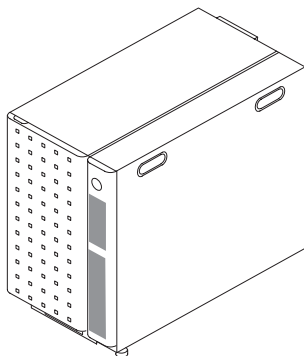
2306872**LCD COLOR
MONITOR

(Pg 23)

2138599-5
EXTERNAL
POWER SUPPLY**2368697-2**LINUX 3-BUTTON
MOUSE

(Pg 21)

NOTE: The following parts are Not Shown.**5132224** OPENSPEED5 OS**5132225** OPENSPEED5 APPLICATION SOFTWARE**5132227** OPENSPEED5 ADVANCED SERVICE SOFTWARE**5132226** OPENSPEED5 RESTRICTED SERVICE SOFTWARE**2396508** SYSTEM SOFTWARE FOR HFO4**5105839** ADVANCED SERVICE SOFTWARE FOR HFO4**5105841** RESTRICTED SERVICE SOFTWARE FOR HFO4***2295019** MONITOR DATA CABLE FOR 18" LCD****2213962-4** POWER SUPPLY FOR NEC2010 MONITOR****2150533-2** CABLE, NEC2010 MONITOR (821/798) OW1A15/OW1A2A3

GOC COMPUTER CABINET FOR HF05:**5117866-22**

HP XW8200

2235744-5

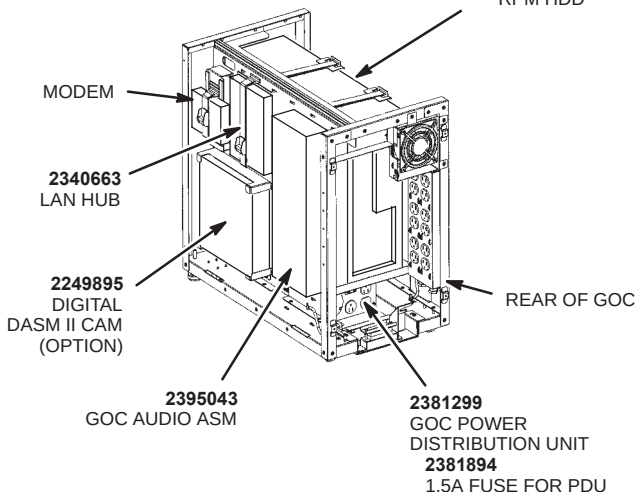
BIT3 BD FOR

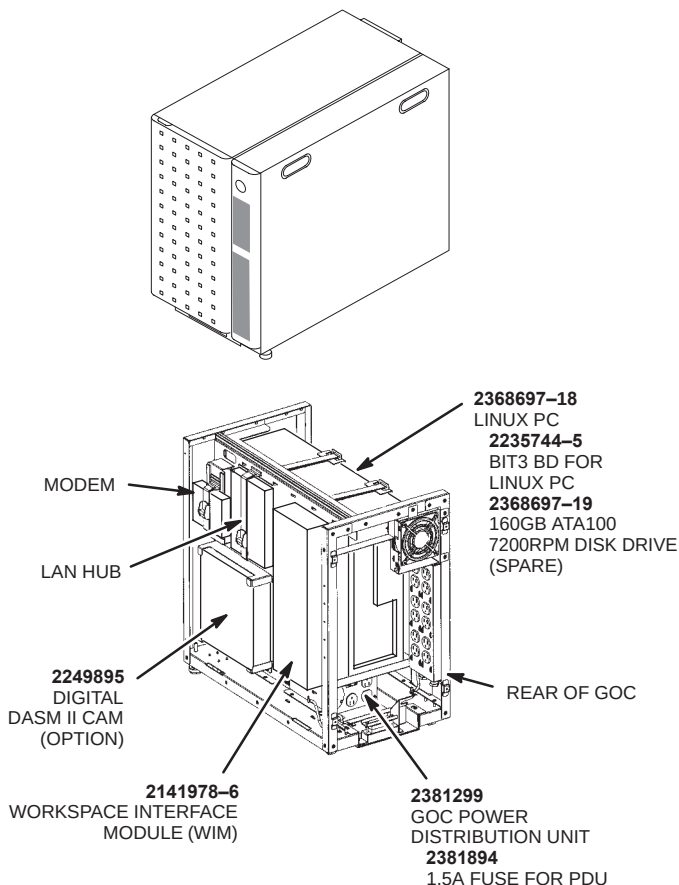
LINUX PC

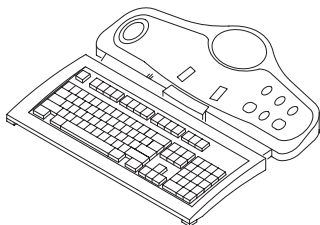
5117866-23

160GB SATA/150 7200

RPM HDD



GOC COMPUTER CABINET FOR HF04:

SCIM:**KEYBOARDS ONLY:**

- 2275756** (English)
- *2275756-2** (French)
- *2275756-3** (German)
- *2275756-4** (Scandinavian)
- *2275756-5** (Italian)
- *2275756-6** (Portuguese)
- *2275756-7** (Spanish)

*** NOTE:**

These types of Keyboard/SCIM Assemblies were not available at the time this book was released, see below for earlier types.

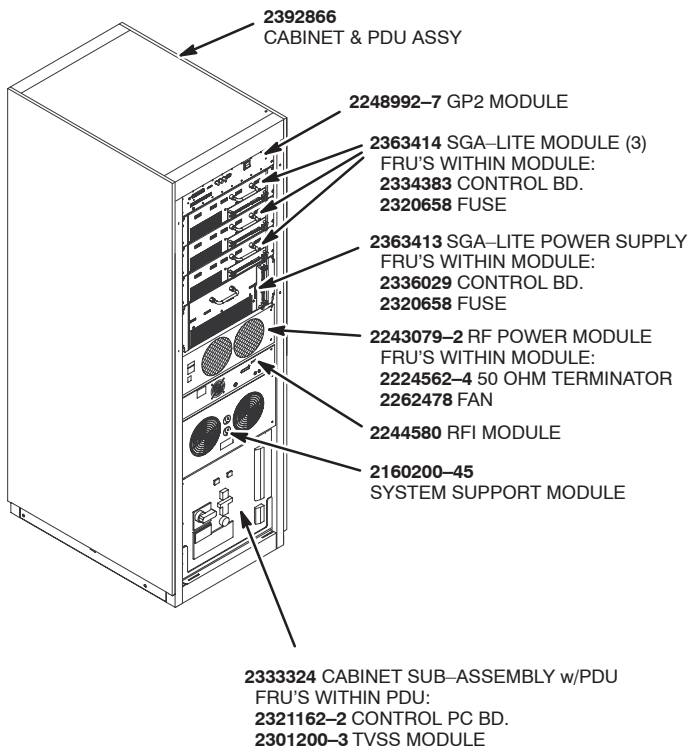
NOTE: The following parts are Not Shown.

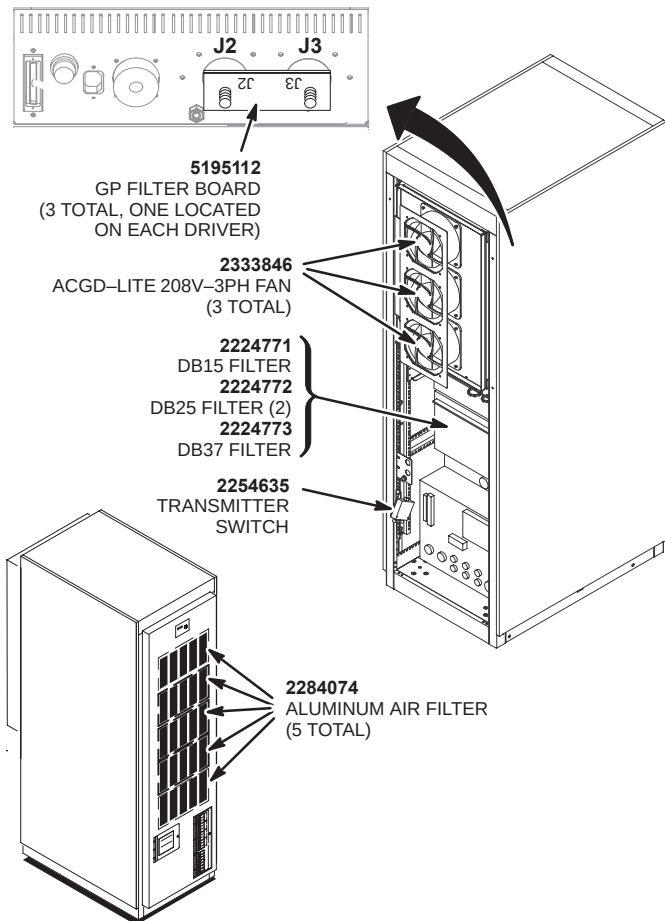
2395735	GOC SCIM-AUDIO CBL (FOR GOC AA)
2262158-9	CABLE, 50 PIN SCIM KEYBOARD(FOR WIM)
2289697-2	SCIM ^W /ENGLISH OVERLAY
2289697-8	MR SCIM GLOBAL OVERLAY SET

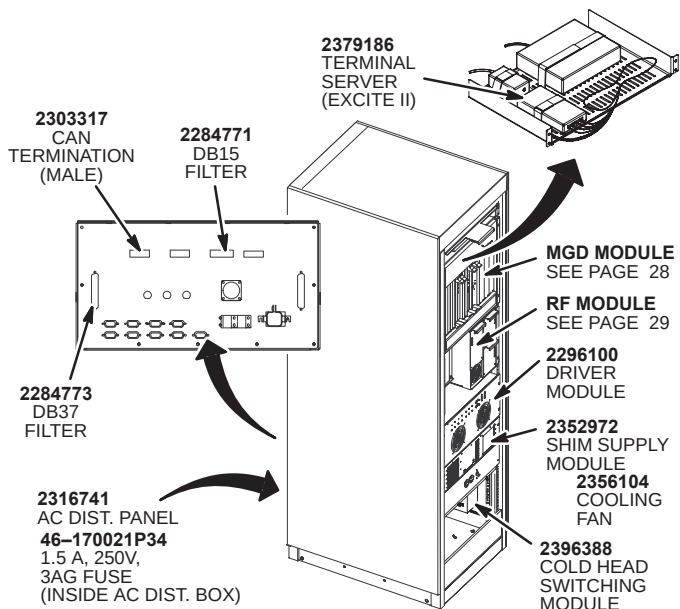
[illegible]

MAIN DISCONNECT PANEL (MDP)

OpenSpeed systems may have an MDP manufactured by either GE Industrial Systems or Ellis & Watts. For information on respective MDP's, refer to Vendor Service Manuals on the Service Methods CD-ROM under Parts tab.

GRFD CABINET





NOTE: The following System Cabinet Options are Not Shown.

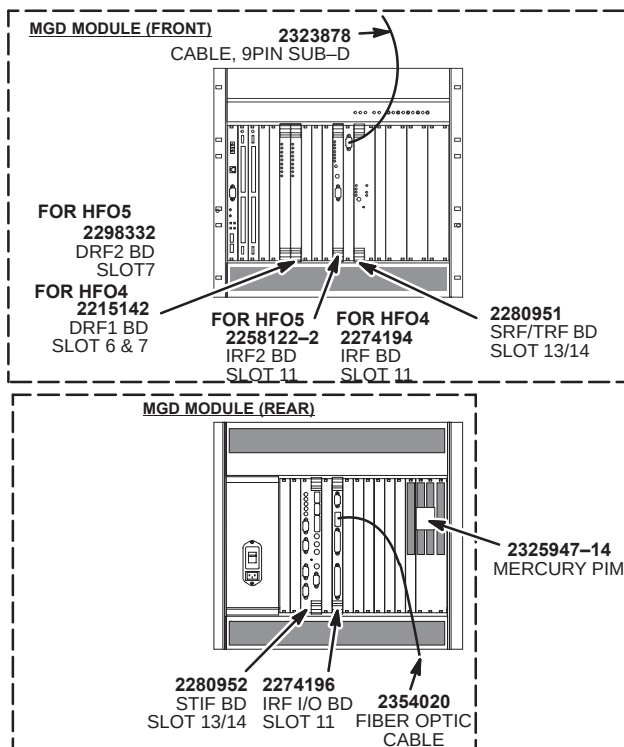
2325947-11	MAVERICK II
2325947-12	MINERVA II MASTER
2325947-13	MINERVA II SLAVE
2325947-14	MERCURY PIM

OPTION CONFIGURATIONS:

REFLEX 100 2325947-11 in Slot 2

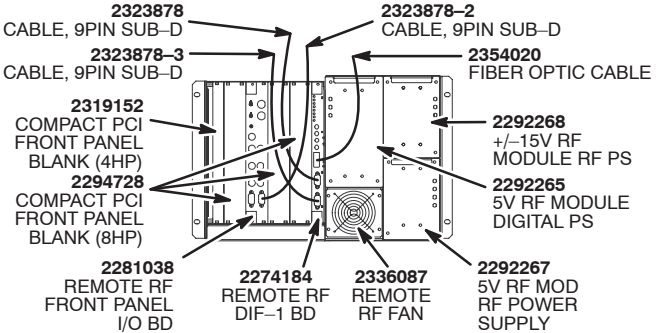
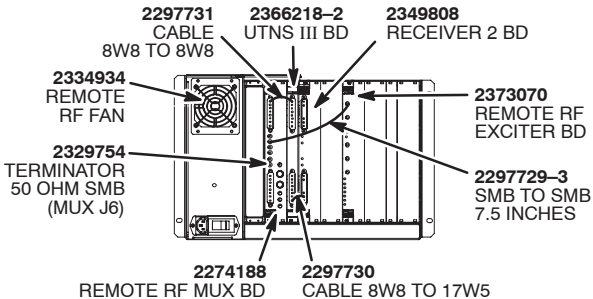
REFLEX 200 2325947-11 in Slots 2 and 3
Use PIM 2325947-14

RELEX 400B 2325947-12 in Slot 2
2325947-13 in Slot 3
Use PIM 2325947-14



NOTE: The following MGD Module parts are Not Shown.

2294300-2	SCP BOARD WITH CAN DAUGHTER BD
2294300-3	AGP BOARD
2294300-4	APS BOARD WITH IT DAUGHTER BD
2294300-5	BRIDGE BOARD
2294300-6	MGD CHASSIS W/BACKPLANE & AC ENTRY (No Boards)
2294300-7	MGD FAN
2294300-8	POWER SUPPLIES
2294300-9	INDICATOR BOARD
2294300-12	SCP2 BOARD

2305884 REMOTE RF MOD ASM (CHASSIS W/MIDPLANE) (FRONT)**2305884 REMOTE RF MOD ASM (CHASSIS W/MIDPLANE) (REAR)**

NOTE: The following Parts/Cables are Not Shown.

2272740-3	8 PORT 10/100 INTEL HUB SERVICE KIT
46-306565P1	CABINET FILTER
2308729	BNC TO SMB CABLE (MUX J8 TO CAB I/F J1)
2308728	8W8 CABLE ASSEMBLY (MUX J5)
2322578	J15/23/25/26 DATA INPUT CABLE (CAB I/F)
2322580	CABLE J18, UNBLANK (CAB I/F)
2322581	CABLE J24, MAGNET MONITOR (CAB I/F)
2322583	CABLE J62 (15 PIN SUB-D) (CAB I/F)
2322586	CABLE J61 (37 PIN SUB-D) (CAB I/F)
2315093	ETHERNET CABLE
2315093-2	CABLE J63 ETHERNET (CAB I/F)
2308729-2	BNC TO SMB CABLE (MUXJ13 TO CAB I/F J2)
2308729-3	BNC TO SMB CABLE (MUXJ14 TO CAB I/F J3)

[illegible]

PHANTOM CART KIT**2274476**

LV SHIM

INNER SPHERE

2274477

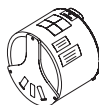
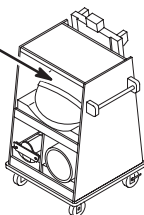
LV SHIM

UPPER HALF

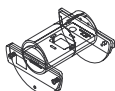
2274479

LV SHIM

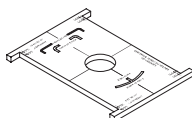
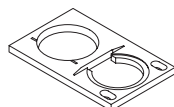
LOWER HALF

**2125244**SPT SHORT LOADER
w/SPHERE ASM

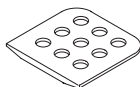
For Field Replacement Order:

2135650-2 – SPT SPHERE**2135652-2** – SPT LOADER**2274999**

DQA ASM

**46-328383P1**UNIVERSAL
PHANTOM HOLDER**2274392**

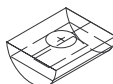
NESTING PLATE

**2282698**

GRAFIDY PLATE

**46-265826G6***

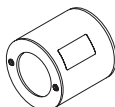
TLT HEAD SPHERE

**2170481***EPI FOAM
POSITIONER**2141454***QUICK SHIM
POSITIONER

* Typically not used in service procedures for OpenSpeed.

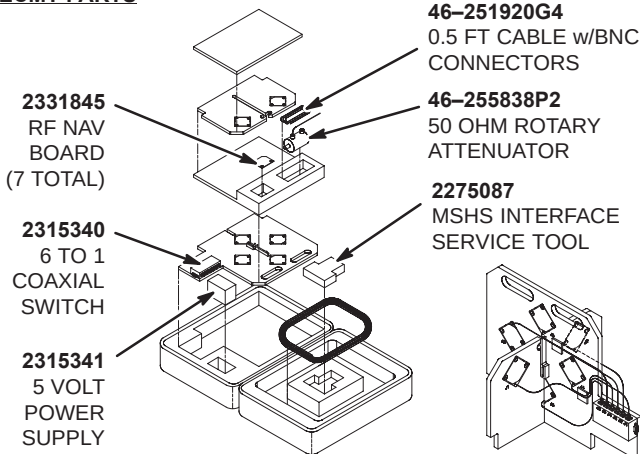
EPI PHANTOM KIT**2258366**

4.38" PHANTOM

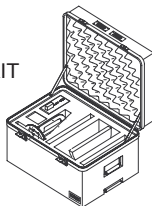
**2350593**

HEAD LOADER

**2357209**EPI POSITIONING
PADS

ECMT PARTS

**2281754,
2281754-2**
SHIM CAMERA KIT

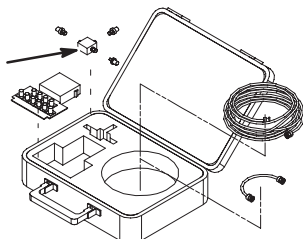


2360581
0.7T M-C RECEIVE TOOL KIT

2364386
M-C RECEIVE
CHAIN HFO2 PATCH

2366015
M-C RECEIVE CHAIN
TROUBLESHOOT TOOL

2360522
BIAS TEE



COILS:

* Service Manuals for Coils available on Service CD-ROM under Parts > Vendor Coils

* Where possible, FRU's which are part of Option Catalogs are shown under the areawhere they would be installed, rather than being listed in this section.

HEAD COIL (LARGE BASEPLATE) M2087H

2274240-7	HEAD ARRAY COIL & BASEPLATE ASM ^{W/} INTREFC CABLE
2274240-2	HEAD ARRAY COIL SYSTEM INTERFACE CABLE
2274240-5	HEAD ARRAY PHANTOM
2274240-8	HEAD ARRAY PHANTOM POSITIONER

HEAD COIL (SMALL BASEPLATE) M2087HA

2299522-2	HEAD ARRAY COIL & BASEPLATE ASM ^{W/} INTREFC CABLE
2274240-2	HEAD ARRAY COIL SYSTEM INTERFACE CABLE
2274240-5	HEAD ARRAY PHANTOM
2274240-8	HEAD ARRAY PHANTOM POSITIONER

**BREAST COIL M2087BR**

2264745-7	BREAST COIL ^{W/} CABLE
2264745-2	BREAST COIL CABLE ONLY
2246360-9	PHANTOM POSITIONER FOR BREAST COIL
46-317586G1	SPHERE PHANTOM

CTL COIL (SPINE) M2087HS

2264744-12	CTL ARRAY ^{W/} CABLE
2264744-2	CTL CABLE ASSEMBLY ^{W/} QD BOX
2264744-11	CTL ARRAY LATCH MECHANICS KIT
2264744-7	CTL ARRAY CT PHANTOM
2264744-8	CTL ARRAY TL PHANTOM

EXTREMITY / FOOT SET M2087E

2264742-2	LARGE KNEE COIL ^{W/} CABLE ASSEMBLY
2264742-14	LARGE KNEE COIL CABLE ASSEMBLY
2264742-8	KNEE COIL PHANTOM (LARGE OR SMALL)
2264742-10	PHANTOM POSITIONER (LARGE & SMALL)
2264742-3	SMALL KNEE AND FOOT COIL ^{W/} CABLE
2264742-5	SMALL KNEE / FOOT COIL CABLE ASSEMBLY
2264742-13	LATCH PARTS (SET OF 2)
2264742-4	FOOT PHANTOM

COILS CONTINUED ON NEXT PAGE.

*COILS CONTINUED FROM PREVIOUS PAGE.***NEUROVACULAR ARRAY COIL M2087NC**

2396079-2	NV ARRAY COIL
2396079-3	HEAD RESTRAINT ASM
2396079-4	PHANTOM POSITIONER
2396079-5	PATIENT COMFORT PAD
2396079-8	RECTANGULAR PHANTOM
2396079-9	ROUND PHANTOM
2396079-10	CABLE
2396079-11	MIRROR ASM

SHOULDER COIL M2087SH

2264743-9	SHOULDER ARRAY COIL ^{W/} CABLE
2264743-2	SHOULDER ARRAY COIL CABLE ASSEMBLY
2264743-5	SHOULDER ARRAY PHANTOM POSITIONER
46-265826G6	HEAD TLT PHANTOM

WRIST COIL M2087WR

2279915-2	PHASED ARRAY WRIST COIL & BASEPLATE ASSEMBLY
2279915-11	PHASED ARRAY WRIST COIL BASEPLATE CABLE
2279915-6	PHASED ARRAY WRIST COIL PHANTOM
2279915-10	PHASED ARRAY WRIST COIL PHANTOM POSITIONER

LARGE BODY FLEX COIL M20032GC

2273180-2	BODY FLEX L COIL ^{W/} CABLE
2148175	BODY FLEX LOCK M1
2148176	BODY FLEX LOCK M2
2169786-2	BODY FLEX LOCK F1
2169787-2	BODY FLEX LOCK F2

MEDIUM BODY FLEX COIL M20022GC

2273181-2	BODY FLEX M COIL ^{W/} CABLE
2148175	BODY FLEX LOCK M1
2148176	BODY FLEX LOCK M2
2169786-2	BODY FLEX LOCK F1
2169787-2	BODY FLEX LOCK F2

OPTIONS:**AP POWER PC W/128MB – DUAL**

2222532	PPC COMPATIBLE MEDCAM BASEBOARD
2222532-3	DUAL PPC MODULE W/64MB (32MB/CE NODE)
2101767-4	MEDCAM VMX 128 MB MEMORY BOARD
2101767-5	MEDCAM VMX FOREPLANE 1-SLOT COVER

[illegible]

NOTES